AI- Driven Phishing Email Detection Using NLP (Natural Language Processing)

Project Report

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1. Introduction

1.1 Problem Statement

, email is one of the most common ways of communication. People use emails for personal work, business, education, and many other purposes. Every day, millions of emails are sent around the world. However, not every email is useful or trustworthy. Many unwanted emails, known as spam emails, are sent to users every day.

Spam emails may contain advertisements, fake offers, phishing links, or harmful content. These emails can waste users' time, fill inboxes, and sometimes even steal personal information. Checking every email manually is difficult because a large number of emails are received every day. Therefore, there is a need for a system that can automatically identify spam emails with good accuracy.

Machine Learning provides a practical solution to this problem. By learning patterns from previously labeled emails, a machine learning model can classify new emails as spam or ham based on their textual content. Instead of checking emails manually, the model can make the filtering process faster and more efficient.

In this project, a spam email detection system has been developed using Natural Language Processing (NLP) and Machine Learning techniques. The dataset contains both spam and ham emails. After cleaning and preparing the data, different machine learning models are trained and evaluated to identify the model that performs best for spam email classification. The main objective of this project is to build a simple and effective system that can classify emails using their text.

1.2 Importance of Fake News Detection

Spam emails can create many problems for users. They can waste time, reduce productivity, and sometimes contain phishing links or harmful attachments that may lead to security risks. As more people depend on email for daily communication, detecting spam emails has become an important task.

An automatic spam email detection system can help users by filtering unwanted emails before they reach the inbox. Such systems can also support businesses, educational institutions, and email service providers by reducing the time and effort needed to manage large numbers of emails.

Natural Language Processing (NLP) plays an important role in spam email detection because machine learning models cannot understand text directly. The text must first be converted into numerical features before it can be used for prediction. In this project, Bag of Words and TF-IDF techniques are used for feature extraction, and multiple machine learning models are trained and compared using the same dataset.

The results of this project show that machine learning techniques can classify spam and ham emails with high accuracy when proper data preprocessing, feature extraction, and model evaluation methods are applied. Therefore, spam email detection is an important application of Artificial Intelligence that helps improve email security and user experience.

2. Dataset Description

2.1 Dataset Source

The dataset used in this project contains spam and legitimate (ham) email messages collected from multiple publicly available email datasets. These datasets were combined to create a single dataset for training and evaluating the machine learning models.

The combined dataset contains email text and a corresponding label that identifies whether an email is spam or legitimate. Before model training, the dataset was cleaned and prepared for further analysis. The email text from different sources was merged into one column named text_combined, while the labels were stored in the label column.

The prepared dataset was then used for text preprocessing, feature extraction, model training, and performance evaluation.

2.2 Dataset Size

The final dataset used in this project contains a total of **82,078 email records** after removing duplicate entries during data preprocessing.

Dataset	Number of Records	Number of Columns
Spam & Legitimate Emails (Combined)	82,078	2

Table 2.1 Dataset Size

The dataset contains one text feature and one target label. The label value **0** represents legitimate (ham) emails, while **1** represents spam emails. This cleaned dataset was used for feature extraction, model training, and model evaluation.

2.3Dataset Features

The dataset contains one input feature and one target label used for machine learning.

Feature	Description
text_combined	Complete email text used for spam detection
label	Target variable (0 = Legitimate Email, 1 = Spam Email)

Table 2.2 Dataset Features

The **text_combined** column contains the complete email content after combining the available email text. This text was cleaned and converted into numerical features using the TF-IDF technique before training the machine learning models.

```
df.head()
```

	text_combined	label
0	hpl nom may 25 2001 see attached file hplno 52...	0
1	nom actual vols 24 th forwarded sabrae zajac h...	0
2	enron actuals march 30 april 1 201 estimated a...	0
3	hpl nom may 30 2001 see attached file hplno 53...	0
4	hpl nom june 1 2001 see attached file hplno 60...	0

Figure 2.1 Preview of the Spam Email Dataset

3. Methodology

3.1 Data Preprocessing

Data preprocessing is an important step in any machine learning project because the quality of the input data directly affects the model's performance. Before training the machine learning models, the dataset was cleaned and prepared to make it suitable for feature extraction and classification.

The project started with a combined dataset containing spam and legitimate email messages. The dataset was first checked for missing values and duplicate records. Duplicate emails were removed to improve data quality and avoid repeated information during model training.

After cleaning the dataset, the email text was preprocessed using basic Natural Language Processing (NLP) techniques. All text was converted to lowercase to maintain consistency. Special characters, punctuation, numbers, and extra spaces were removed. Common English stop words were also removed to keep only meaningful words. These preprocessing steps helped reduce noise in the dataset and improved the quality of the input text.

The cleaned email text was stored in a new column named **clean_text**, which was later used for feature extraction and machine learning model training.

```
df["clean_text"].head(10)

0    hpl nom may see attached file hplno xls hplno xls
1    nom actual vols th forwarded sabrae zajac hou ...
2    enron actuals march april estimated actuals ma...
3    hpl nom may see attached file hplno xls hplno xls
4    hpl nom june see attached file hplno xls hplno...
5    hpl nom may see attached file hplno xls hplno xls
6    tried get fancy address came back forwarded la...
7    hpl noms february see attached file hplo xls h...
8    fw pooling contract template original message ...
9    hpl nom march see attached file hplo xls hplo xls
Name: clean_text, dtype: str
```

Figure 3.1 Data Preprocessing Steps

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset before training the machine learning models. This step helped examine the distribution of spam and legitimate emails and identify important patterns in the dataset.

Different visualizations were created to understand the class distribution and compare the length of spam and legitimate emails. These graphs helped verify that the dataset was suitable for machine learning and provided useful insights before feature extraction.

The EDA results showed that both spam and legitimate emails were well represented in the dataset. The analysis also showed that email length varies between the two classes, which can provide useful information for spam email classification.

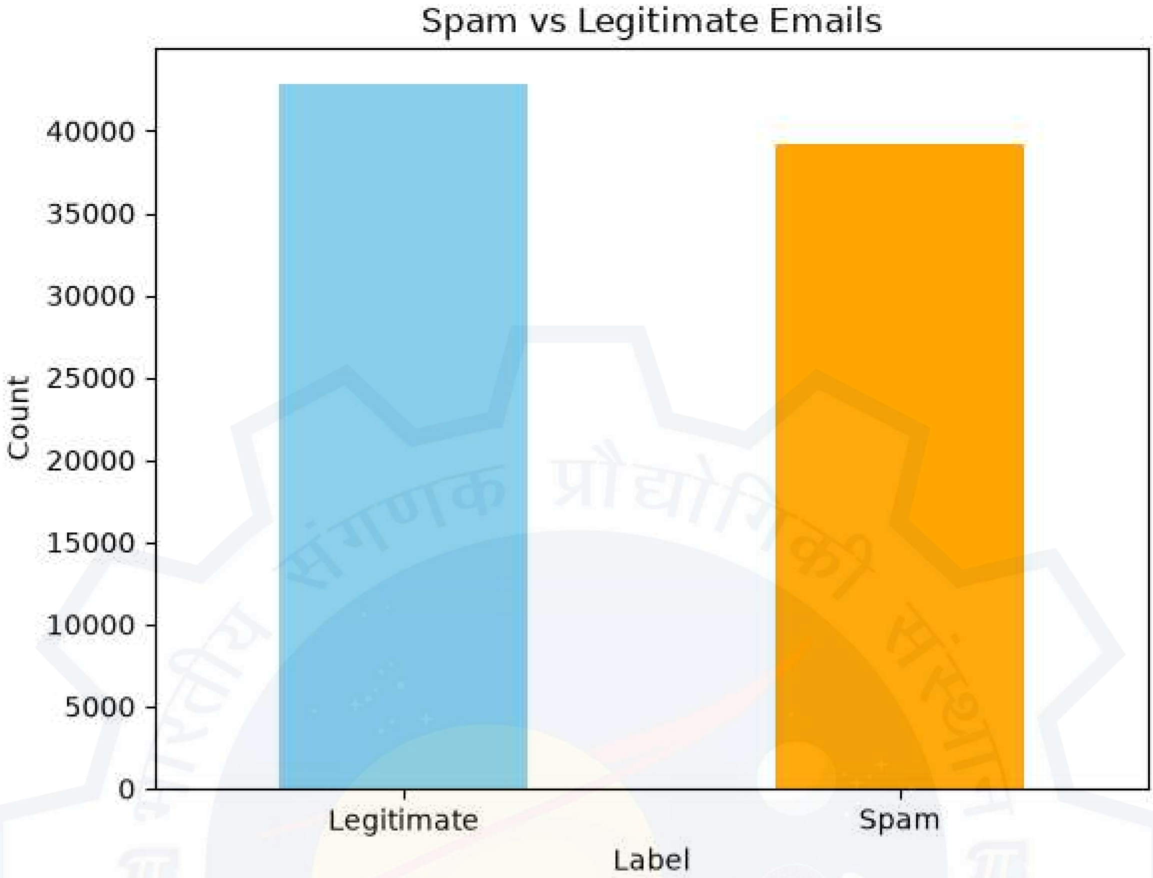

Figure 3.2 Distribution of Spam and Legitimate Emails

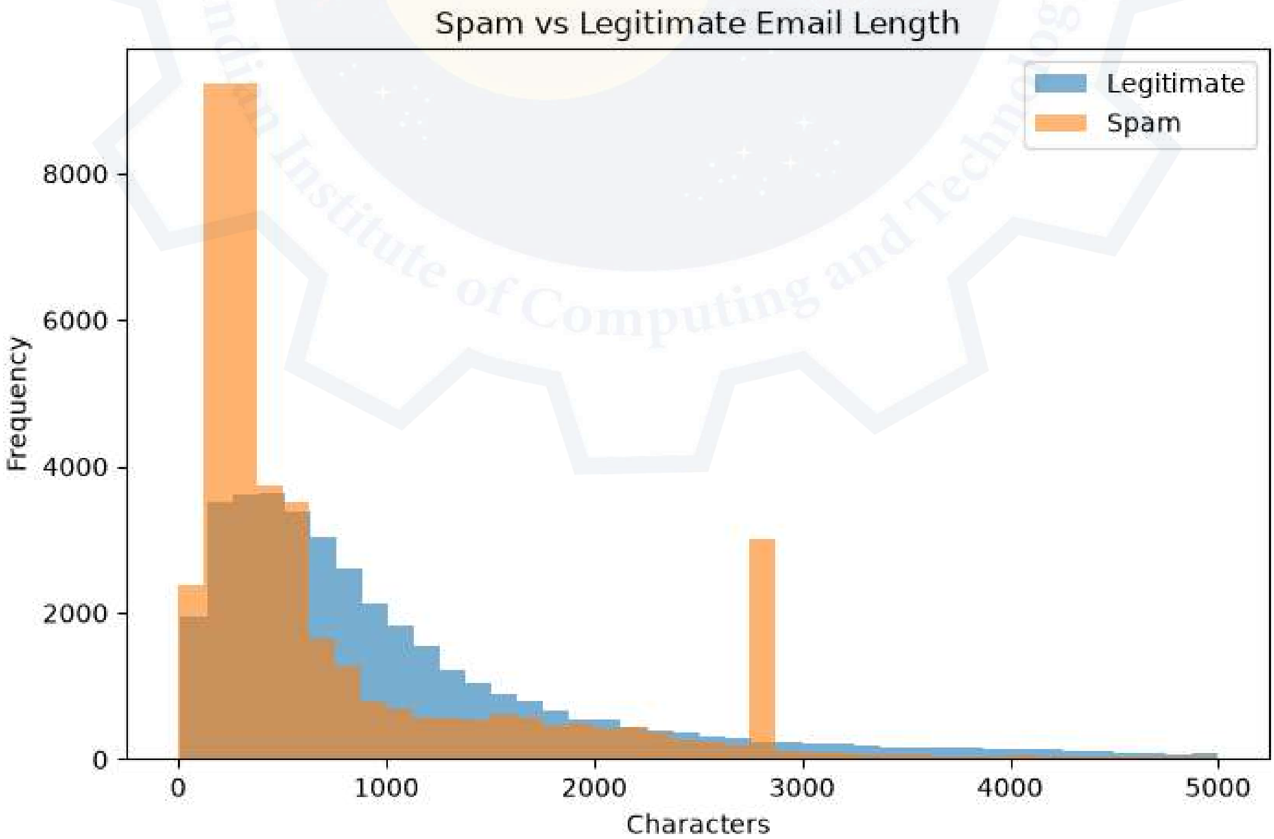

Figure 3.3 Comparison of Spam and Legitimate Email Length

3.3 Feature Extraction

Machine learning models cannot understand text directly. Therefore, the cleaned email text was converted into numerical features before training the models. In this project, the Term Frequency-Inverse Document Frequency (TF-IDF) technique was used for feature extraction.

TF-IDF converts each email into a numerical feature vector based on the importance of words in the dataset. Words that appear frequently in a particular email but are less common across other emails receive higher importance scores. This helps the machine learning models focus on meaningful words while reducing the effect of very common words.

In this project, the TF-IDF vectorizer was created with 5000 maximum features. The transformed feature matrix was then used as the input for all machine learning models.

```
top20 = feature_df.head(20)
top20
```

	Feature	Importance
360	aug	0.035718
4960	wrote	0.031664
1475	enron	0.025418
3343	pm	0.015712
4484	thanks	0.014189
2572	list	0.013990
4319	subject	0.009234
2667	mailing	0.009121
771	click	0.008547
2874	money	0.008336
2163	im	0.008190
334	attached	0.008150
1095	date	0.007957
1720	file	0.007805
4688	university	0.007679
2550	life	0.007329
3094	opensuse	0.007117
2119	http	0.006934
660	cc	0.006741
4862	wed	0.006264

Figure 3.4 TF-IDF Feature Matrix

3.4 Train-Test Split

After feature extraction, the dataset was divided into training and testing datasets. The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate their performance on unseen email data.

An 80:20 split was used in this project, where 80% of the data was used for training and the remaining 20% was used for testing. A fixed random_state = 42 was used to ensure that the data split remained the same every time the project was executed.

Using the same train-test split for all machine learning models ensured a fair comparison of their performance.

```
    print("X_train :", X_train.shape)
    print("X_test  :", X_test.shape)
    print("y_train :", y_train.shape)
    print("y_test  :", y_test.shape)

X_train : (65662, 5000)
X_test  : (16416, 5000)
y_train : (65662,)
y_test  : (16416,)
```

Figure 3.5 Train-Test Split Output

3.5 Machine Learning Algorithms

Four machine learning algorithms were used in this project to classify emails as spam or legitimate. All models were trained using the same training dataset and tested using the same testing dataset. Using the same dataset and train-test split for every model ensured that the comparison was fair and the performance of each algorithm could be evaluated under the same conditions.

The selected algorithms represent different machine learning approaches and are widely used for email and text classification tasks. Each model learns patterns from the TF-IDF feature vectors and predicts whether an email belongs to the spam or legitimate category.

- **Multinomial Naive Bayes** is a probabilistic machine learning algorithm that is simple, fast, and widely used for text classification problems. It performs well with TF-IDF features and is commonly used for spam email detection.

- **Logistic Regression** is a linear classification algorithm that learns the relationship between the input features and the target class. It is widely used for binary classification tasks because of its simplicity and good prediction performance.

- **Random Forest** is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It helps reduce overfitting and provides reliable performance on large text datasets.

- **Neural Network (MLP Classifier)** was used to learn more complex patterns from the TF-IDF feature vectors. It consists of interconnected layers that help identify relationships within the email text and improve classification performance.

After training, the performance of all four models was evaluated using **accuracy, precision, recall, F1-score, classification report, and confusion matrix**. These evaluation metrics were used to compare the models and identify the best-performing algorithm for spam email detection.

4. Results

4.1 Model Performance

After training the machine learning models, their performance was evaluated using the testing dataset. Each model was tested using the same train-test split to ensure a fair comparison. The evaluation was based on standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix.

Model	Accuracy (%)
Multinomial Naive Bayes	95.90
Logistic Regression	98.29
Random Forest	98.44
MLP Classifier	98.43

Table 4.1 Model Accuracy Comparison

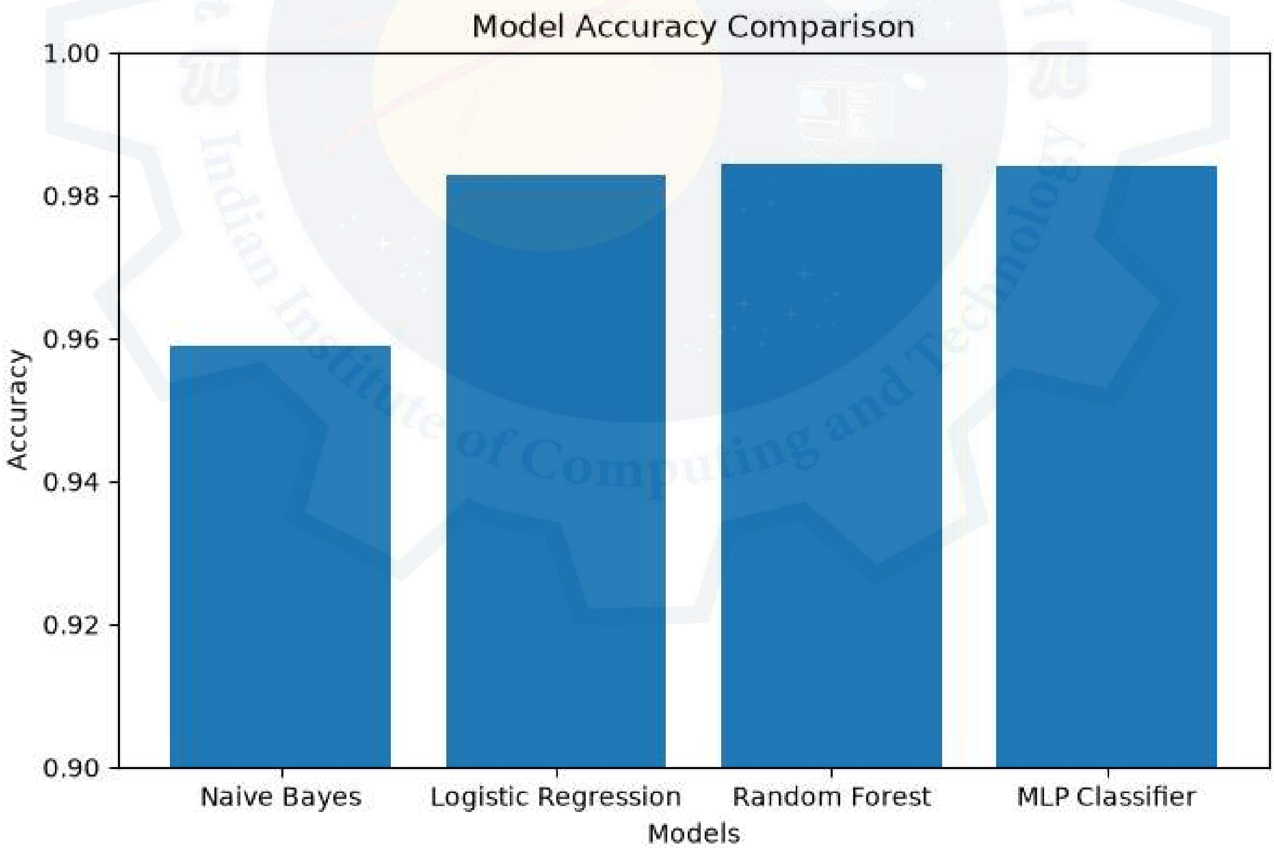

Figure 4.1 Model Accuracy Comparison

The results show that all models achieved good performance on the dataset. However, the accuracy of each model was different because every algorithm uses a different learning approach.

4.2 Classification Report

In addition to accuracy, precision, recall, and F1-score were used to evaluate the classification performance of the models. These metrics provide a better understanding of how correctly each model identifies real and fake news articles.

- **Precision** measures how many predicted news articles are correctly classified.
- **Recall** measures how many actual news articles are correctly identified.
- **F1-score** provides a balanced measure of precision and recall.

These metrics help evaluate the reliability of the machine learning models beyond overall accuracy.

4.3 Confusion Matrix

A confusion matrix was used to visualize the classification results of each machine learning model. It shows the number of correctly classified and incorrectly classified emails for both spam and legitimate classes.

The diagonal values of the confusion matrix represent correct predictions, while the remaining values represent incorrect predictions. A higher number of correct predictions indicates better model performance.

4.4 Model Comparison

The performance of all four machine learning models was compared using the same evaluation metrics and the same testing dataset. This comparison helped identify the strengths of each algorithm and determine which model performed best for phishing email detection.

Among all the models, Random Forest achieved the highest accuracy of 98.44%, followed closely by the MLP Classifier with 98.43% and Logistic Regression with 98.29%. Multinomial Naive Bayes also performed well with an accuracy of 95.90%, but its performance was slightly lower than the other models.

The comparison shows that ensemble learning and neural network approaches produced better results than the probabilistic approach used in this project.

4.5 Best Performing Model

Based on the evaluation results, Random Forest was selected as the best-performing model for this project. It achieved an accuracy of 98.44%, which was the highest among all the machine learning algorithms evaluated.

The model also produced high precision, recall, and F1-score, indicating that it classified both phishing and legitimate emails effectively. Its strong performance demonstrates that ensemble learning is well suited for phishing email detection tasks.

Although the MLP Classifier and Logistic Regression also achieved excellent results, Random Forest provided the best overall performance on the testing dataset and was selected as the final model for this project.

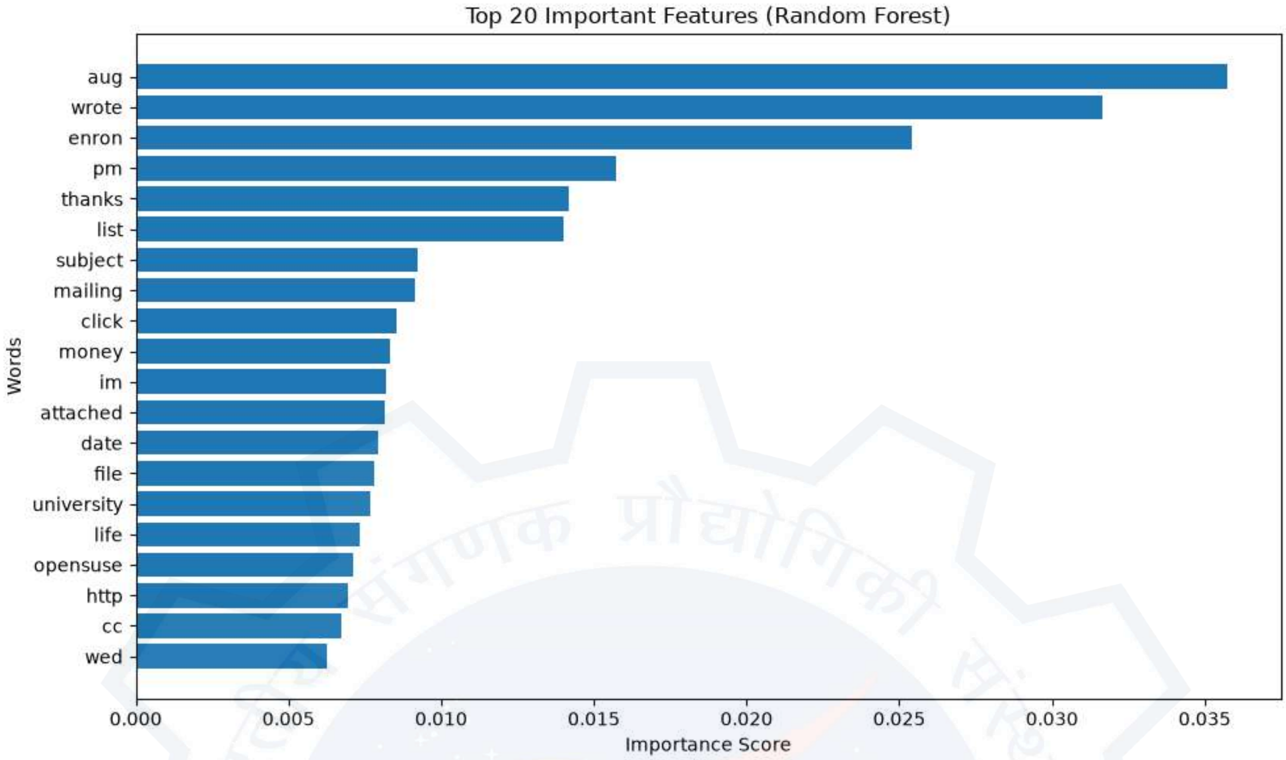

Figure 4.2 Top 20 Important Features

5. Discussion

5.1 Interpretation of Results

The experimental results show that machine learning techniques can effectively classify emails as phishing or legitimate using Natural Language Processing (NLP) and TF-IDF feature extraction.

All four machine learning models achieved high classification accuracy, although their performance varied because each algorithm uses a different learning approach.

Among all the models, Random Forest achieved the highest accuracy and provided the most reliable predictions. MLP Classifier and Logistic Regression also performed very well, showing that they are suitable for phishing email detection tasks. Multinomial Naive Bayes achieved slightly lower accuracy but still produced reliable classification results.

The comparison of different models shows that selecting an appropriate algorithm plays an important role in improving the performance of a phishing email detection system. Overall, the models successfully learned meaningful patterns from email text and were able to classify phishing and legitimate emails with high accuracy.

5.2 Strengths of the Project

The project has several strengths that contribute to its overall performance and reliability.

- The dataset contains both phishing and legitimate email samples, providing suitable data for model training and evaluation.
- Proper data preprocessing improved the quality of the email text before model training.

- TF-IDF feature extraction successfully converted email text into meaningful numerical features.
- Multiple machine learning algorithms were implemented and compared using the same dataset and evaluation process.
- Standard evaluation metrics such as accuracy, precision, recall, F1-score, classification report, and confusion matrix were used to compare the models.
- The project achieved high classification accuracy, with **Random Forest** providing the best overall performance (**98.44%**).

These strengths demonstrate that the developed system can effectively classify phishing emails and can serve as a useful example of applying Natural Language Processing and Machine Learning to cybersecurity problems.

5.3 Limitations of the Project

Although the project achieved excellent results, it also has a few limitations.

- The models were trained and tested using a single dataset, so additional datasets may improve their ability to generalize to different phishing email patterns.
- The system classifies emails mainly based on textual content and does not consider other information such as sender reputation or real-time email metadata.
- The project focuses only on English-language email messages.
- Advanced deep learning models such as LSTM, BERT, or Transformer-based models were not implemented.
- The performance of the system depends on the quality of the training dataset and the preprocessing steps applied before model training.

These limitations provide opportunities for improving the system in future work.

5.4 Future Scope

The performance and usability of the phishing email detection system can be improved by extending the project in several ways.

Future work may include using larger and more diverse email datasets collected from different sources to improve the model's ability to detect new phishing techniques. Advanced deep learning models such as LSTM, BERT, and Transformer-based architectures can also be explored to improve classification performance.

The system can be integrated into email services, web applications, or browser extensions to provide real-time phishing email detection. Additional email features such as sender information, URLs, attachments, and domain analysis can also be included to improve detection accuracy.

These improvements can help develop a more accurate, reliable, and practical phishing email detection system for real-world cybersecurity applications.

6. Conclusion

The objective of this project was to develop a machine learning system that can classify emails as phishing or legitimate using Natural Language Processing (NLP) and text classification techniques. To achieve this objective, the email dataset was collected, preprocessed, converted into numerical features using TF-IDF, and used to train multiple machine learning models.

Four machine learning algorithms, namely Multinomial Naive Bayes, Logistic Regression, Random Forest, and MLP Classifier, were implemented and evaluated using the same training and testing datasets. Their performance was compared using standard evaluation metrics, including accuracy, precision, recall, F1-score, classification report, and confusion matrix.

The experimental results showed that all four models achieved high classification performance. Among them, Random Forest achieved the highest accuracy of 98.44%, making it the best-performing model in this project. The comparison of different algorithms demonstrated that machine learning techniques can effectively identify patterns in email text and accurately distinguish phishing emails from legitimate emails.

Overall, this project demonstrates the practical application of Machine Learning and Natural Language Processing in cybersecurity. The developed system provides a simple and effective approach for automated phishing email detection and can serve as a foundation for future improvements and real-world applications.

7. Appendix

The appendix provides supporting information related to the implementation of the fake news detection system. It includes sample data, important code snippets, model outputs, and additional results that support the findings presented in the main report.

Appendix A: Sample Dataset

Figure A.1 Sample of the Email Dataset

Appendix B: Important Code Snippets

The following code sections were used during the implementation of the project:

- Import Required Libraries
- Dataset Loading
- Data Preprocessing
- Text Cleaning
- TF-IDF Feature Extraction
- Train-Test Split
- Model Training

- Model Evaluation
- Custom Email Prediction

Only the key code snippets should be included instead of the complete notebook.

Appendix C: Model Evaluation Outputs

This appendix contains additional outputs generated during model evaluation.

Appendix C includes the classification reports generated for:

- Multinomial Naive Bayes
- Logistic Regression
- Random Forest
- MLP Classifier

	precision	recall	f1-score	support
0	0.94	0.98	0.96	7910
1	0.98	0.94	0.96	8506
accuracy			0.96	16416
macro avg	0.96	0.96	0.96	16416
weighted avg	0.96	0.96	0.96	16416

Figure A.2 Multinomial Naive Bayes Classification Report

	precision	recall	f1-score	support
0	0.98	0.98	0.98	7910
1	0.98	0.98	0.98	8506
accuracy			0.98	16416
macro avg	0.98	0.98	0.98	16416
weighted avg	0.98	0.98	0.98	16416

Figure A.3 Logistic Regression Classification Report

	precision	recall	f1-score	support
0	0.98	0.99	0.98	7910
1	0.99	0.98	0.98	8506
accuracy			0.98	16416
macro avg	0.98	0.98	0.98	16416
weighted avg	0.98	0.98	0.98	16416

Figure A.4 Random Forest Classification Report

	precision	recall	f1-score	support
0	0.98	0.99	0.98	7910
1	0.99	0.98	0.98	8506
accuracy			0.98	16416
macro avg	0.98	0.98	0.98	16416
weighted avg	0.98	0.98	0.98	16416

Figure A.5 MLP Classifier Classification Report

Appendix D: Confusion Matrices

Appendix D presents the confusion matrices generated for all five machine learning models. These matrices provide a detailed visualization of the classification performance of each model by comparing actual and predicted classes.

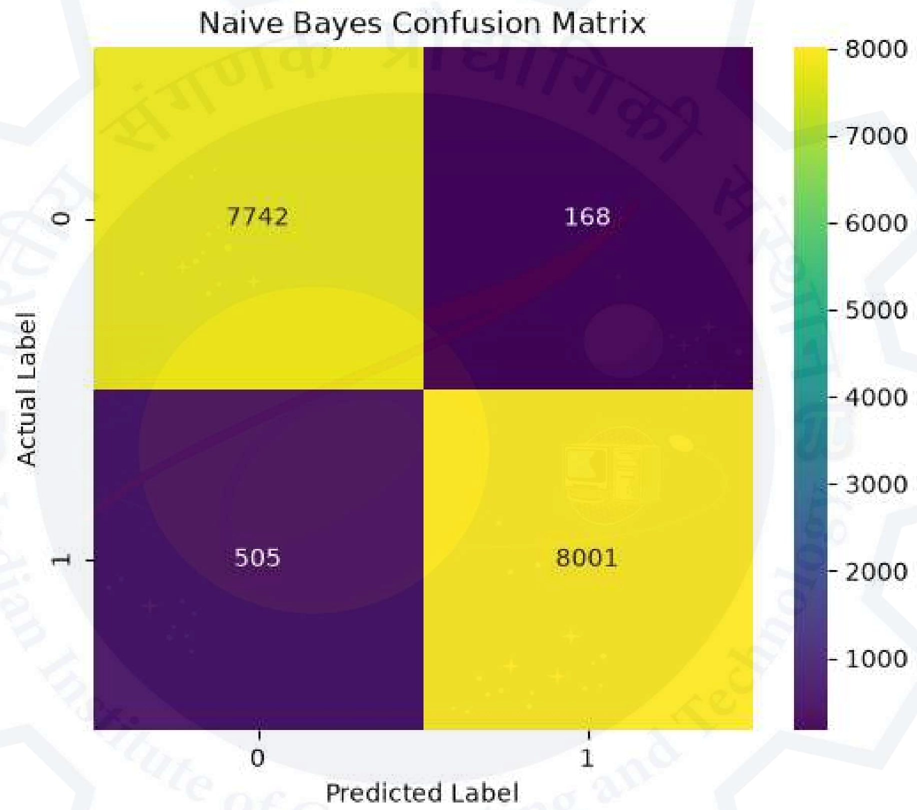

Figure A.6 Multinomial Naive Bayes Confusion Matrix

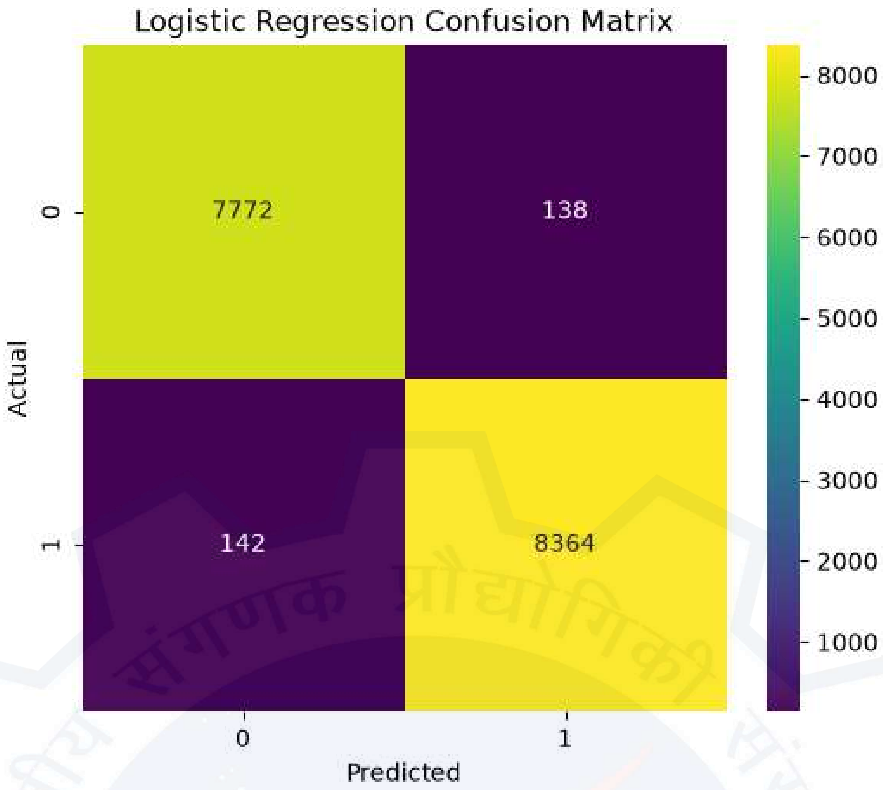

Figure A.7 Logistic Regression Confusion Matrix

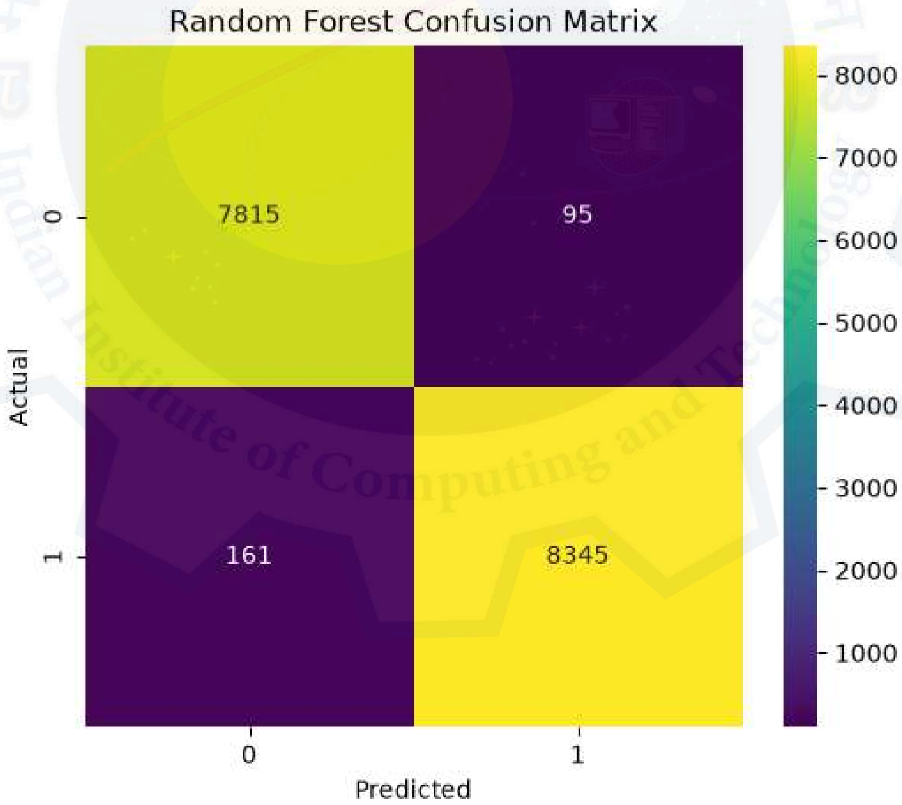

Figure A.8 Random Forest Confusion Matrix

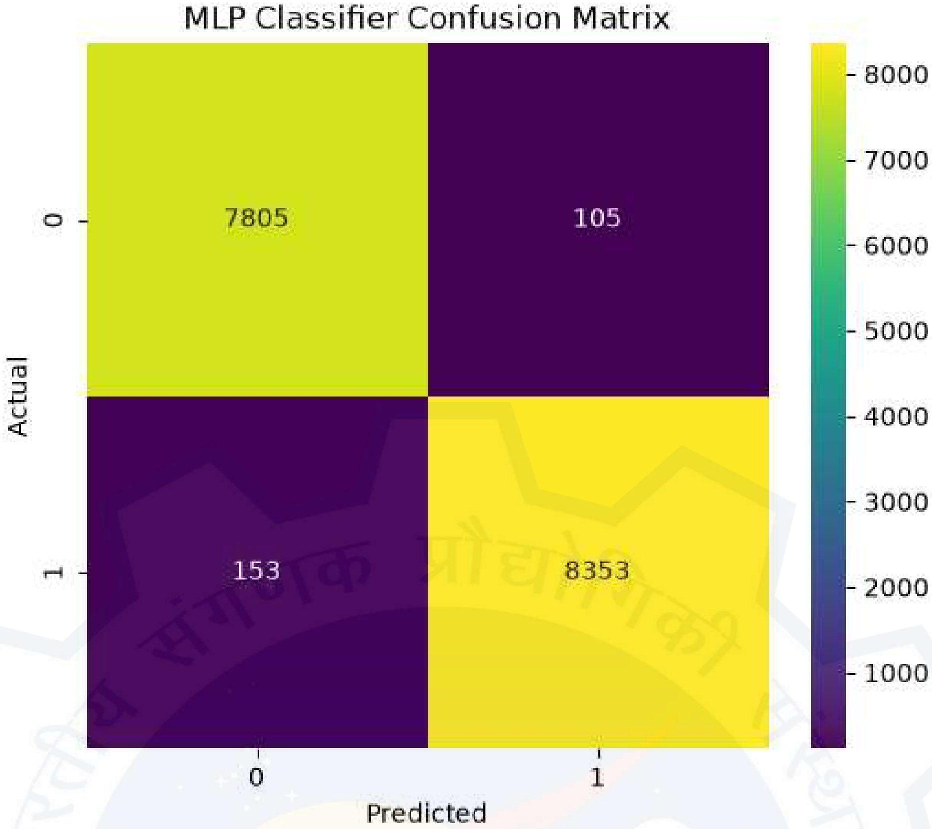

Figure A.9 MLP Classifier Confusion Matrix

Appendix E: Software Tools and Libraries

The following software tools and Python libraries were used to implement the project.

Software	Purpose
Python	Programming Language
Jupyter Notebook	Development Environment

Table A.2 Software Used

Library	Purpose
pandas	Data Handling
numpy	Numerical Computing
matplotlib	Visualization
re	Text Cleaning
nltk	Stopword Removal
scikit-learn	Machine Learning
joblib	Save Models

Table A.3 Python Libraries

Appendix F: Model Accuracy Summary

	Model	Accuracy
0	Naive Bayes	0.959003
1	Logistic Regression	0.982943
2	Random Forest	0.984405
3	MLP Classifier	0.984284

Table A.4 Model Accuracy Summary

Summary

Among all the machine learning algorithms, Random Forest achieved the highest accuracy of 98.44%, making it the best-performing model for phishing email detection. The MLP Classifier and Logistic Regression also achieved excellent performance with accuracies above 98%, while Multinomial Naive Bayes provided good performance with an accuracy of 95.90%.